

Hengyu Liang

Computer Science and Technology
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EDUCATION BACKGROUND

- **University of Science and Technology of China (USTC)** Sep. 2023 - Present
M.S. in Computer Science and Technology
GPA: 3.79/4.3
Hefei, China
- **University of Science and Technology of China (USTC)** Sep. 2019 - Jun. 2023
B.E. in Computer Science and Technology
GPA: 3.97/4.3; Grade Ranking: 3/254
Hefei, China
Member of Hua Xia Talent Program (consisting of 27 Talents from over 1,800 freshmen in Class 2019 at USTC)

PUBLICATION

- Yuming Xu, **Hengyu Liang**, Jin Li, et al., “SPFresh: Incremental In-Place Update for Billion-Scale Vector Search,” in Proceedings of the 29th Symposium on Operating Systems Principles (SOSP ’23), Association for Computing Machinery, New York, NY, USA, 2023, pp. 545–561.
<https://doi.org/10.1145/3600006.3613166>

INTERNSHIP EXPERIENCE

- **Microsoft Research Asia(MSRA)**
Research member supervised by Ms Qi Chen, a Principal Researcher at MSRA
Beijing, China
- **Project One:** Jul. 2022 - Jun. 2023
Incremental In-Place Update for Billion-Scale Vector Search (SPFresh)
 - Improved the RocksDB storage backend by tuning configurations, adopting the latest io_uring I/O API, experimenting with a KV separation architecture, etc.
 - Further introduced SPDK and designed an efficient KV storage engine to meet the project’s specific performance requirements
 - Completed the overall performance evaluation of the SPFresh system, which included experimental design, data analysis, and result visualization
 - Revealed that compared to the existing DiskANN system, our SPFresh reduced peak memory usage from 1100GB to less than 20GB, decreased the number of front-end CPU cores from 32 to 2, lowered tail latency by 2.4 times, and achieved a stable online update throughput for billions of vectors
 - As the second author, I co-composed and published a research paper titled SPFresh: Incremental In-Place Update for Billion-Scale Vector Search on the SOSP 2023.
- **Project Two:** Apr. - Jun. 2022
Performance Evaluation and System Optimization of Vector Search Systems
 - Defined core testing objects, i.e., graph-based methods (e.g., HNSW), tree-based methods (e.g., Annoy), cluster-based methods (e.g., IVF series), and hash-based methods (e.g., LSH)
 - Designed performance testing metrics, developed experimental scripts, addressed code differences across various repositories, and wrote automated testing scripts to systematically test the aforementioned methods on standard datasets (e.g., SIFT, GIST)
 - Collected testing results and compiled a detailed performance evaluation report, presenting each method’s applicability and limitations
 - Further proposed potential optimization strategies, including parameter tuning, index structure optimization, and algorithm implementation improvements

RESEARCH EXPERIENCE

- **Advanced Data Systems Laboratory (ADSL)** <https://adsl.ustc.edu.cn/main.htm>
Research Assistant to Prof. Cheng Li USTC
Sept. 2023 - Present
Optimization of Filtered Vector Search System for Billion-Scale Data
 - Analyzed the existing Filtered DiskANN system and identified its design flaws and performance bottlenecks, especially in low-selectivity query scenarios
 - Proposed an innovative hybrid index structure that combines tree-based and graph-based indexes
 - Optimized the index construction workflow to reduce memory usage and improve construction efficiency
 - Implemented the unique hybrid-index disk-based ANNS system called SPFilter
 - Developed an automated experiment script system to test the SPFilter system and identified its superior disk-based filtering and search performance when processing billion-scale data
 - At present, I am working on ongoing optimization and further validation of the SPFilter's performance across diverse selection conditions
- **Project Two:** Aug. 2021 - Jun. 2022
Research on Metadata Load Balancing in the Distributed Storage System CephFS
 - Preprocessed time series data, conducted feature engineering, built and trained multiple time series prediction models, and selected appropriate ones based on prediction accuracy and computational cost
 - Integrated PyTorch into CephFS runtime, independently fixing compatibility via compiler version adjustments and dependency coordination for stable co-operation
 - Identified high inference latency of neural networks on CPUs and proposed optimization strategies, such as periodic predictions, service separation, and using lightweight models to address the issue

ACADEMIC PROJECT EXPERIENCE AT USTC

- **Research on Storage Sandboxes Based on the Linux Kernel** Apr. - Jul. 2021
Group leader supervised by Prof. Kai Xing from the School of Computer Science and Technology, USTC
 - Conducted a systematic survey of various kernel sandbox technologies in Linux, mainly including namespace isolation mechanisms, seccomp system call filtering, BPF (Berkeley Packet Filter) kernel-mode extension programs, etc.
 - Analyzed the applicable scenarios and performance costs of these technologies in storage isolation and I/O security control
 - Designed and implemented a simple storage sandbox based on the Linux kernel according to the analysis results.
 - Revealed that the implemented storage sandbox can achieve flexible file system access restrictions and permission management for specific processes

TEACHING ASSISTANT EXPERIENCE AT USTC

- **Teaching Assistant** Sept. 2021 – Jan. 2022
Analog and Digital Circuits
 - Responsible for grading and collecting assignments, analyzing common errors for targeted recitation sessions, and organizing focused review lectures on core concepts prior to exams.

HONORS & AWARDS

China National Scholarship for Excellence in Academics (top 1%)	Dec. 2020
Excellent Student Scholarship – Gold (top 3%)	Jun. 2021

SKILLS & HOBBIES

Computer skills: C/C++ (8 yrs), Python (5 yrs), L^AT_EX(5 yrs), bash (5 yrs)
Hobbies: Photography(Member of the USTC Omnimedia Center), Traveling, etc.

Scholastic Record

Reg.NO: PB1906123		Name: Lianz Hengyu		Gender: Male		Date of Birth: 2001-05-23				
Enrl Date: 2019-09-01		Dep Date: 2023-07-01		E.S: 4						
School: Department of Computer Science and Technology										
Major: Computer Science and Technology		All Curriculum GPA: 3.97				All Curriculum Weighted Average Score: 91.98				
Term	Course Title	Gr.	Hrs.	Cr.	Term	Course Title	Gr	Hrs	Cr.	
2019FA	Military Theory	Pass	40	2	2020SP	C Language Programming II	94	80	3	
2019FA	College English Viewing, Listening and Speaking (Band 2)	92	20	1	2020SP	Mechanics and Thermal Physics	100	80	4	
2019FA	Calculus for Functions of a Single Variable	95	120	6	2020SP	College Physics Experiment I	100	60	1.5	
2019FA	Ideological and Moral Cultivation and Basics of Law	82	60	3	2020SP	Electromagnetism C	100	60	3	
2019FA	College English for Reading & Writing (Intermediate)	88	40	2	2020SP	Algebraic Structure	90	60	3	
2019FA	Linear Algebra B1	94	80	4	2020SP	Freshman Seminar	A+	20	1	
2019FA	Computer Programming A	95	100	4	2020SP	Physical Fitness I	Pass	40	1	
2019FA	Basic Sports	B+	40	1	2020SP	College English Viewing, Listening and Speaking (Band 3)	85	20	1	
2020SP	Calculus for Functions of Several Variables	94	120	6	2020SP	College English for Reading & Writing (Advanced)	88	40	2	
2020SP	An Outline of Modern and Contemporary Chinese History	86	60	3						
2020FA	Graph Theory	87	60	3	2021SP	Practice on Chinese Important Thoughts	Pass	80	2	
2020FA	College Physics - Comprehensive Experimentation	94	60	1.5	2021SP	Computer Organization	96	100	4	
2020FA	Optics and Atomic Physics	95	80	4	2021SP	The Beauty and Joy of Computing I	A	60	2	
2020FA	Data Structures	89	100	4	2021SP	Operating Systems (H)	97	100	4	
2020FA	Analog and Digital Circuits	93	80	4	2021SP	Introduction to Chinese Important Thoughts	90	60	3	
2020FA	College English Viewing, Listening and Speaking (Band 4)	95	20	1	2021SP	Mathematical Logic	97	60	3	
2020FA	Introduction to Computer Systems(H)	97	100	4	2021SP	A Guide to Japanese Art	A	40	2	
2020FA	Fundamentals of Marxism	85	60	3	2021SP	College English Viewing, Listening and Speaking (Band 5)	87	20	1	
2020FA	Syllabus of Digital Logic Lab	89	40	1	2021SP	Operations Research	95	40	2	
2020FA	Badminton II	B+	40	1	2021SP	Badminton I(club)	B+	40	1	
2020FA	Function of Complex Variable B	93	40	2	2021SP	Probability and Statistics B	83	60	3	
2021SP	Equations of Mathematical Physics B	92	40	2						
2021FA	Situation and Policy	Pass	40	2	2022SP	Fundamental of Artificial Intelligence	86	80	3.5	
2021FA	Stochastic Processes B	83	40	2	2022SP	Computational Methods B	82	40	2	
2021FA	Introduction to Data Science	A-	40	2	2022SP	An Introduction to Database System	86	90	3.5	
2021FA	Computer Networks	95	80	3.5	2022SP	Introduction to Software Engineering	83	40	2	
2021FA	Foundations of Algorithms	90	90	3.5	2022SP	Computer Graphics	99	60	3	
2021FA	Principles and Techniques of Compiler(H)	90	100	4	2022SP	Parallel Computing	84	60	2.5	
2021FA	Design Methods of Embedded System	96	60	2.5	2022SP	Multimedia and Its Application	93	80	3.5	
2022SP	Computer Architecture	88	90	3.5						
2022FA	Scientific English	89	20	1	2022FA	Advances in Computer Science Researches	95	30	1.5	
2022FA	Advanced Software Engineering	94	180	6						
Graduation Thesis		Name: Index reordering optimization based on IVF-Index				Supervisor: Li Cheng		A+	8	2023SP

Note: (H) represents the curriculum of Honors; FA: Fall SP: Spring SU: Summer
Grades on the pass/fail system do not count toward GPA and weighted average score

GPA Calculation:

Centesimal Grade:

Letter Grade:

Point Value:

100~95	94~90	89~85	84~82	81~78	77~75	74~72	71~68	67~65	64	63~61	60
A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-
4.3	4	3.7	3.3	3	2.7	2.3	2	1.7	1.5	1.3	1

$$\text{GPA} = \frac{(\text{Course Credit} * \text{Course GP})}{\text{Course Credit}}$$